

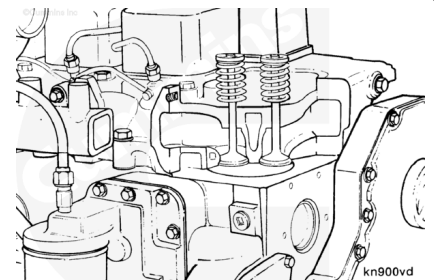
Trainings ▾

General Information

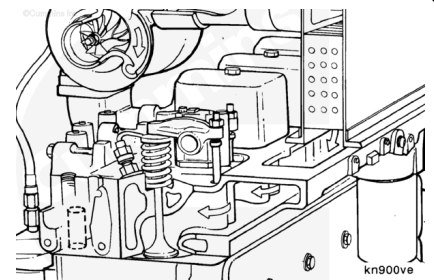
B3.9, B5.9, and B4.5 Engines

Cylinder Head and Valve Train

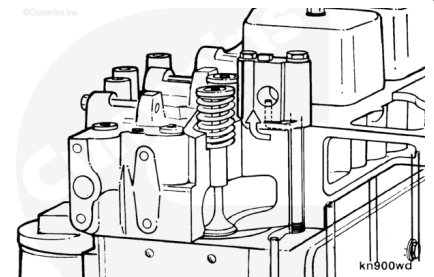
The cylinder head is a one-piece, cross flow design with two valves per cylinder. The head has integrally cast valve guides and hardened valve seat surfaces that can be repaired in a machine shop using the appropriate service parts.



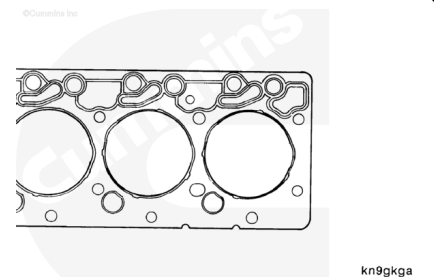
The cylinder head has a cast intake manifold, fuel filter head, thermostat housing and an internal water bypass. The injectors are mounted in the head for direct injection into the cylinders. Fuel filter head is remote for engines with in-line fuel injection pump.



Separate pedestals for each cylinder are to support and route oil to the rocker levers.



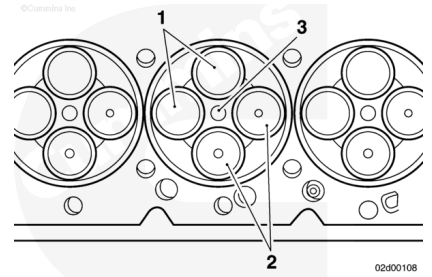
The cylinder head gasket is a specialized metal design with a printed o-seal on both sides around the water holes. An embossment in the gasket seals the cylinder bores. As discussed in the Cooling System, the gasket also provides orifices to control coolant flow.



B4.5 RGT Engines

Cylinder Head and Valve Train

The cylinder head is a one-piece, cross flow design with four valves per cylinder. The cylinder head has an integral intake manifold, thermostat housing and an internal water bypass. The four valve per cylinder design allows for a center injector in the cylinder head (3).



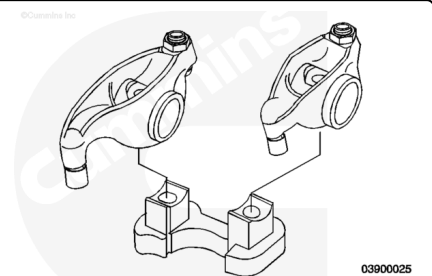
The valves (1 and 2) are made of heat resistant steel, and have chrome plated stems to prevent scuffing. The intake and exhaust valves are both similar in head diameter and overall length, but they have unique face angles which makes them non-interchangeable. The valves are distinguished by unique dimples on the exhaust valve head (2).

The exhaust valve springs are made from high strength, chrome silicon steel. The exhaust valve springs are also exhaust brake compatible.

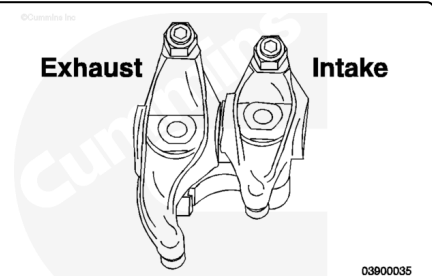
The head has integrally cast valve guides which are **not** serviceable. The valve seats are inserts that can be serviced if damaged.

The exhaust and intake rocker levers are mounted on a common pedestal, but rotate on separate shafts.

Oil is supplied through a drilling in the cylinder head through the pedestals to supply oil to the rocker lever shafts, socket and an adjusting screw.

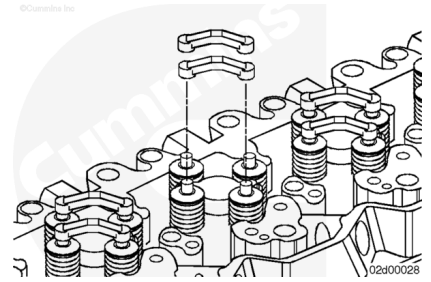


The exhaust and intake rocker levers are **not** interchangeable and **must** be located as shown.



With the four valve per cylinder design, the rocker lever sockets do **not** directly contact the valve stem. With one rocker lever being required to operate two valves, a bridge, or crosshead, is needed to connect both intake and exhaust valves.

The socket on the rocker lever contacts the crosshead, which in turn contacts both corresponding valve stems.



The cylinder head gasket is a specialized metal design with a printed o-seal on both sides around the water holes. An embossment in the gasket seals the cylinder bores. As discussed in the Cooling System, the gasket also provides orifices to control coolant flow.

No specific head gasket with an increased thickness is available for cylinder head or block combustion deck resurfacing. If the combustion deck can **not** be resurfaced such that the piston protrusion, valve depth and injector protrusion specifications be maintained, the cylinder head and/or cylinder block **must** be replaced.

